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Moving to sustainable servitization: categorisation and analysis of underlying approaches

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HIGHLIGHTS

- Lack of conceptual clarity concerning the heterogeneity of manufacturers' approaches to sustainable servitization.
- Four categories of approaches based on service orientation and business model scheme.
- Different resource-saving tactics and, thus, different resource requirements.
- Resource requirements and customer expectations as core rigidities hampering the adoption of certain approaches.
- Guidance to develop a sustainable servitization orientation and activate the necessary resources.

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ABSTRACT

As rising environmental concerns create the need to embed sustainability more centrally into business propositions, the emerging research on sustainable servitization is looking into the possibility for manufacturing companies to combine service-led growth with the generation of environmental value. Existing characterisations, however, interpret sustainable servitization as a generic strategy, failing to recognise the heterogeneity of approaches that it may encompass. This paper develops a taxonomy that – based on service orientation (lifecycle management or operational efficiency) and business model scheme (ownership-based or access-based) – classifies manufacturers' approaches to sustainable servitization into four distinct categories, which differ fundamentally in their sustainability tactics and underlying resource requirements. We also investigate how these categories resonate with practice by analysing the cases of three manufacturers of industrial equipment that have embarked on a sustainable servitization journey. By proposing our taxonomy and unpacking each of the four approaches, we provide granular insights into what it takes for a manufacturer to integrate a sustainability perspective into its servitization strategy. This granular analysis is important as it adds depth and nuance to the understanding of sustainable servitization implementation, which would otherwise be overly simplistic and inaccurate. We also identify that resource requirements and customers' strong focus on economic benefits may introduce core rigidities and act as barriers to the widespread adoption of certain approaches. For managers, the study may help think through how to enhance and evolve service businesses in ways that contribute to environmental sustainability, and what resources to activate to support a sustainable servitization orientation over time.

1. Introduction

Starting from the late 1980s, we have witnessed an increasing importance of services in the growth strategies of many manufacturing companies (Vandermerwe and Erixon, 2023). Today, a wide range of manufacturers from different geographies and industries have embraced servitization (Heirati et al., 2023); that is, a transformation process by which a company progressively integrates a variety of services into its product-based value propositions and market offerings as a way to generate extra revenues and secure/improve its competitive position (Kowalkowski et al., 2017). As a case in point, research by Bain & Co. reveals that some machinery and equipment manufacturers currently generate more than 50 percent of revenue and 100 percent of profit through services (Bain & Company, 2022). Similarly, according to a

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McKinsey study, medical product manufacturers on average achieve a 50 percent increase in enterprise value for each additional percentage point by which their service sales grow over product sales (McKinsey & Company, 2022). Despite the traditional rationale for servitization being based on financial, strategic and marketing arguments (e.g., Mathieu, 2001a; Gebauer et al., 2006), recent research has started to recognise that the trend towards servitization may provide a solid foundation for more sustainable production and consumption patterns (Rabetino et al., 2024). The service logic embedded in servitization creates substantial opportunities to improve the use and efficiency of resources (e.g., materials, energy) over the lifecycle of products (Kohtamäki et al., 2024), for example by increasing product longevity and end-of-life recovery, thereby promoting emission reduction and sufficiency (Stabler et al., 2024). Hence, the term 'sustainable servitization' or 'green servitization' has been coined to describe the link between servitization and sustainability.[1]

So far, however, literature and research have mainly treated sustainable servitization as an empirical phenomenon without a coherent view or theoretical conceptualisation of the various ways in which manufacturers may integrate sustainability in their servitization strategies. The primary focus has been on providing empirical evidence about the positive impact of sustainable servitization adoption on companies' environmental (e.g., Kanatlı and Karaer, 2022; Kumar et al., 2024), financial (e.g., Kohtamäki et al., 2024) and other performance outcomes (e.g., Upadhayay et al., 2024), and in turn demonstrating that extending research in this field has both theoretical and practical relevance. In contrast, the concept of sustainable servitization is poorly defined and articulated (Rabetino et al., 2024). Researchers have adopted different perspectives as to the nature, rationale and boundaries of the core concept, which has led to a variety of different characterisations and operationalizations. Nevertheless, existing characterisations interpret sustainable servitization as a generic strategy, failing to recognise the heterogeneity of approaches that it may encompass (Zhou et al., 2024). Each approach carries its own resource-saving tactics, operational models and practices, and therefore its implementation requires a specific array of operand (e.g., physical) and operant (e.g., knowledge) resources. The lack of conceptual clarity regarding the heterogeneity of possible approaches to sustainable servitization, along with the neglect of the associated resource requirements, represents an important gap in the literature. Against this background, the objective of this paper is to propose a taxonomy that categorises and describes, from a conceptual and resource perspective, manufacturers' approaches to sustainable servitization. In particular, we address the following research questions.

1. How can manufacturers' approaches to sustainable servitization be categorised to capture their heterogeneity?
2. What are the implications of such heterogeneity in terms of types of resources required for the implementation of each approach?

In line with the mainstream literature, we interpret servitization as the addition of value-added services to the market offerings of product manufacturers (Kowalkowski et al., 2017) and, accordingly, we align to the view that sustainable servitization means that a manufacturer implements a portfolio of sustainable services – i.e., services that embody a sustainability-related value proposition – as part of its servitization strategy (Opazo-Basáez et al., 2018; Marić and Opazo-Basáez, 2019; Benedettini, 2022; Upadhayay et al., 2024; Gebauer et al., 2025). Moreover, following Bigdeli et al. (2018) and Heirati et al. (2023), we regard the categories of service offerings (and ultimately services) that are applicable to manufacturing companies as servitization approaches. Accordingly, we equate a manufacturer's pursuit of a particular sustainable servitization approach with the offering of a given category of sustainable services. Our taxonomy of possible approaches to sustainable servitization is, indeed, a taxonomy of manufacturers' sustainable services.

We investigate the research questions above using a two-stage research approach. In the first stage, we apply the iterative method proposed by (Nickerson et al., 2013) to develop a taxonomy that groups manufacturers' sustainable services into four categories based on service orientation (lifecycle management or operational efficiency) and business model scheme (ownership-based or access-based). The second stage adopts a case approach to provide an empirical assessment of the different sustainable servitization approaches in our taxonomy and of the associated resource requirements. Our research advances the understanding of sustainable servitization and its implementation. It can help managers of manufacturing companies think through how to extend and enhance service portfolios in ways that contribute to environmental sustainability, or how to evolve sustainable service offerings over time. To appropriately structure sustainable servitization initiatives, these managers need a clear picture of the mechanisms underpinning the generation of environmental value with different services, as well as of the settings that are necessary to enable their successful provision. Seizing sustainable servitization opportunities necessitates activating specialised competences and resources, while also delivering cost savings to customers. Such resource requirements and customer expectations may introduce core rigidities that act as barriers to the widespread adoption of certain approaches.

The paper starts with a description of the conceptual background, followed by the methodology, development of the taxonomy and empirical illustration. Finally, theoretical and managerial implications are discussed, and the study limitations are outlined.

2. Conceptual background

2.1. Sustainable servitization

For a long time, research on servitization has concentrated on economic efficacy and competitive advantage. Product-based companies can use services to develop new revenue streams and create unique customer value, so the common rationale for their move towards services has been based on seizing strategic, financial, and marketing opportunities (Mathieu, 2001a; Gebauer et al., 2006). Driven by mounting pressures to accelerate the transition to a circular economy (CE) and reach the climate goals politically endorsed by the international community (Zhang et al., 2022), the latest research on servitization has expanded its focus beyond profitability and competitiveness to include environmental sustainability. Scholars have started to rethink the role of servitization as a means of more responsible patterns of production and consumption (Doni et al., 2019; Zhou et al., 2024). In particular, an emerging stream of literature, described as sustainable servitization, is looking into the opportunity for manufacturing firms to generate environmental value for their customers and society via the service dimension.

So far, the accounts of this literature have been primarily concerned with examining empirically or analytically the link between sustainable servitization and environmental outcomes. Indeed, even though a wealth of research in the adjacent field of Product-Service Systems (PSS) has extensively theorised the potential of product-service integration to drive eco-efficiency and circularity, actual evidence is lacking and a positive effect is not guaranteed (Kjaer et al., 2019; Marić and Opazo-Basáez, 2019; Roman et al., 2023). Some studies confirm a general positive influence of sustainable servitization on environmental performance (e.g., Oyelakin and Johl, 2022; Johl et al., 2024; Kumar et al., 2024), whereas others report contingent effects. For example, Kanatlı

[1] Although the concept of sustainability also integrates social and economic dimensions, the term sustainable servitization is commonly used to denote a specific focus on environmental sustainability (or 'green'). Therefore, sustainable servitization and green servitization are used as synonyms in the literature. To avoid confusion from using multiple terms to describe the same construct, we use the term sustainable servitization from now on.

and Karaer (2022) find that the achievement of environmental benefits hinges on product-related costs, customer heterogeneity, and market composition. Zhang et al. (2022) identify high service improvement capabilities and low marginal production costs as necessary conditions for the environmental superiority of sustainable servitization over 'pure product' models. In another study, Kolagar (2024) conclude that the most effective configurational strategies to enhance the sustainable performance of industrial customers resort to outcome orientation, Artificial Intelligence (AI)-driven optimisation, and ecosystem orchestration.

Other researchers have turned their attention to the financial impact of sustainable servitization, namely the question of whether environmental benefits may align with manufacturers' economic incentives. These scholars suggest that such alignment is not always achievable (e.g., Kanatlı and Karaer, 2022) and provide evidence of paradoxical tensions between servitization and sustainability that may negatively affect profitability (e.g., Kohtamäki et al., 2024). At the same time, empirical studies detect positive profit effects for specific levels of sustainable servitization (Kohtamäki et al., 2024). Nevertheless, studies drawing upon analytical modelling link win-win outcomes in environmental impact and profitability to adequate levels of product use impact, operating costs, and similarity of customer segments (Örsdemir et al., 2019); or find that the viability of aligning environmental and economic benefits depends on product production costs and pooling efficacy (Agrawal and Bellos, 2017). These results are indicative of the complex mechanisms that appear to underlie the link between sustainability emphasis and financial performance in a large array of prior studies (Kohtamäki et al., 2024; Vaillant and Lafuente, 2024).

Another relevant research stream is concerned with the exploration of the possibilities offered by digital technologies (e.g., sensors, connectivity, AI) and their value in enabling or advancing sustainable servitization by automating and augmenting data-driven actions and decisions (cf. Sjödin et al., 2023). Digital capabilities have been identified as a key antecedent of sustainable servitization (Kumar et al., 2024; Zhou et al., 2024; Chowdhury et al., 2025; Gebauer et al., 2025), as a primary factor that unlocks the benefits of servitization for enhanced environmental outcomes (Paiola et al., 2021; Vaillant and Lafuente, 2024; Kolagar et al., 2024), and as a fundamental enabler of synergies in service provision between environmental and economic sustainability (González Chávez et al., 2024). Regarding economic results in particular, Opazo-Basáez et al. (2018) draw the conclusion that sustainable servitization allows manufacturers to achieve productivity gains only if combined with implementation of digital services. Following the same line of inquiry, Upadhayay et al. (2024) provide evidence that digital and sustainable servitization may complement each other also in shaping companies' innovative performance.

Table 1
Representative definitions of sustainable servitization or green servitization.

Source	Concept[a]	Definition	Operationalisation
Opazo-Basáez et al. (2018)	Green Servitization	Business strategy that seeks to achieve corporate goals for sustainability through provision of green services	Binary variable coded as one if the company has a secondary sector presence in NAICS sector 56 (Administrative support and waste management and remediation services) and/or 811 (Repair and maintenance), zero otherwise
Marić and Opazo-Basáez (2019)	Green Servitization	Business strategy that seeks to achieve corporate goals for sustainability and guides the design, development and implementation of green services from strategic to operational level management	n.a.
Paiola et al. (2021)	Green Servitization	Strategic approach in which corporate business models are designed not only for the achievement of value creation and profits but also to consider and fix some environmental issues	n.a.
Benedettini (2022)	Green Servitization	Servitization strategies of manufacturing companies that are based on the offering of green services; that is, services that are explicitly aimed at helping achieve ecological sustainability objectives	n.a.
Oyelakin and Johl (2022)	Green Servitization	Process of infusing environmentally friendly service solutions into tangible products that are capable of reducing waste, emissions, and fossil fuel energy, including remediation of sites and restoration, repair and maintenance management, water conservation, and recycling of raw materials	Four items representing: (i) green products; (ii) green internal competencies; (iii) green maintenance; (iv) green digital technology
Johl et al. (2024)	Green Servitization	Environmentally friendly service provision	Same as Oyelakin and Johl (2022)
Kohtamäki et al. (2024)	Sustainable Servitization	The interaction between a strategic focus on sustainability (including choices, practices, and business models) and servitization	"sustainab*" + servitization/PSS-related keywords in annual reports
Kumar et al. (2024)	Green Servitization	Business strategy that employs green services to assist firms in meeting their sustainability objectives	Same as Oyelakin and Johl (2022)
Upadhayay et al. (2024)	Green Servitization	Operational approach that prioritises environmentally conscious service-based solutions over traditional production processes	Same as Opazo-Basáez et al. (2018)
Zhou et al. (2024)	Sustainable Servitization	An organizational transformation process via service introduction and advancement that aims to redefine sustainability-related value propositions according to changes in sustainability requirements. It represents a strategic decision and a long-term commitment, benefits from advancements in digital technologies, and typically results in introducing a new business model embedded with an emphasis on continuous improvement	n.a.
Gebauer et al. (2025)	Sustainable Servitization	Service offerings [that] can provide value to customers while also supporting environmental sustainability and resource efficiency	n.a.
Chowdhury et al. (2025)	Green Servitization	Transformation of traditional product-centric business models into service-oriented ones that prioritize environmentally friendly practices to achieve sustainable business performance	Six items representing: (i) use of sustainable materials in service activities; (ii) service operations optimized for energy efficiency; (iii) waste reduction and recycling initiatives for service processes; (iv) service innovation for environmental friendliness; (v) employees trained in environmentally sustainable service practices; (vi) services designed with environmental impact in mind

[a] Sustainable servitization is also referred to as 'green servitization' in the literature. Therefore, included in the table are papers that have provided a definition for either sustainable servitization or green servitization.

Broadly, the extant research demonstrates the theoretical and practical relevance of analysing the strategic restructuring of manufacturing companies towards service orientation from a sustainability perspective. However, researchers tend to adopt idiosyncratic definitions of sustainable servitization that fit the needs of their studies but that are difficult to reconcile with each other. Table 1 shows some of these definitions. Nevertheless, some authors do not refer to an explicit definition but implicitly assume that sustainable servitization indicates the offering of product-as-a-service arrangements (e.g., Agrawal and Bellos, 2017; Örsdemir et al., 2019; Kanatlı and Karaer, 2022; Zhang et al., 2022; Sun et al., 2025). While definitional ambiguity is to be expected in the nascent stage of any new idea of general usefulness (Gladwin et al., 1995), a more concerning issue is that existing characterisations interpret sustainable servitization as a generic strategy, failing to recognise the heterogeneity of approaches that it may encompass. Some scholars have acknowledged that manufacturers' servitization strategies can address sustainability in multiple ways (Spring and Araujo, 2017; Benedettini, 2022; Kohtamäki et al., 2024) and at various levels of the waste hierarchy (Stabler et al., 2024); however, there is a lack of descriptive research that classifies sustainable servitization approaches into distinct categories and provides insights into their heterogeneous nature. Just as the development of general servitization theory has benefited from the effort directed in early research to providing taxonomies of servitization strategies (Kowalkowski and Ulaga, 2024), the same would occur for sustainable servitization theory. Moreover, different approaches to sustainable servitization materialise in different resource-saving tactics, operational models and practices, and thus resort to specific arrays of operand (e.g., physical) and operant (e.g., knowledge) resources. These varying resource requirements imply that some approaches have higher barriers to implementation than others, and thus less likelihood to be widely adopted. In this respect, advancing the dialogue on sustainable servitization necessitates that the specific resource requirements associated with each approach are explicitly identified.

3. Research method

3.1. Stage 1

The study followed a two-stage research design. The first stage focused on formulating our taxonomy of manufacturers' sustainable services. For the development of this taxonomy, we relied on the iterative method proposed by Nickerson et al. (2013) (Fig. 1), which guides researchers in identifying a set of dimensions, each consisting of a set of characteristics, that effectively describe the objects in a specific domain of interest. Though originally conceived for the field of information systems, where taxonomy building had largely been ad-hoc, it has proved to be applicable to non-information systems business disciplines, such as innovation (Karimikia et al., 2022), operations and technology (e.g., Puschel et al., 2022), and general management (e.g., Van Giffen et al., 2022).

As displayed in Fig. 1, the first step of Nickerson et al. (2013)'s taxonomy development method is the definition of a meta-characteristic. In line with the purpose of helping researchers and managers understand the nature and heterogeneity of sustainable servitization approaches, we identified 'service properties that impact the generation of sustainable (environmental) value' as the meta-characteristic for our taxonomy. At every iteration, we ensured that the dimensions and characteristics in the taxonomy related to this meta-characteristic. Then, we defined the objective and subjective conditions that determined when to terminate the iterative procedure. Similar to many prior studies, we opted for the following objective conditions suggested by Nickerson et al. (2013): (i) all objects (manufacturers' sustainable services) or a representative sample of objects have been examined; (ii) at least one object is classified under every characteristic of every dimension; (iii) no dimensions or characteristics were added/split/merged in the last iteration; (iv) every dimension is unique; (v) every characteristic is unique within its dimension. While the last condition allows to claim mutual exclusivity

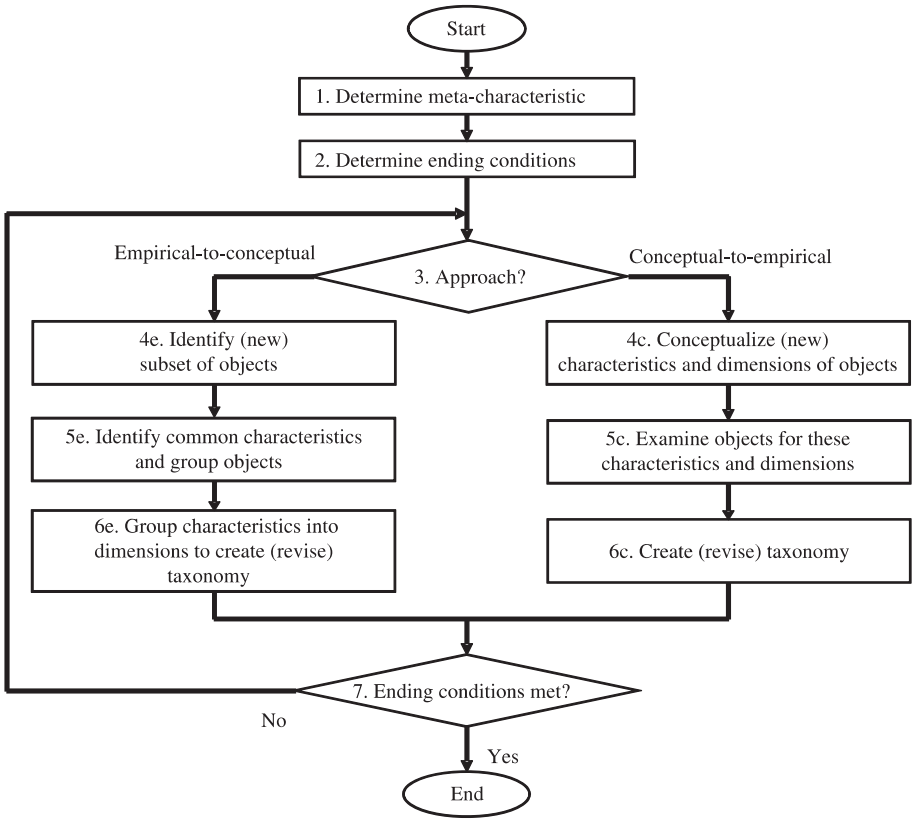

Fig. 1. The taxonomy development method (reproduced with permission: Nickerson et al. (2013)).

among the taxonomy's characteristics, in order to satisfy the requirement of collective exhaustivity, we introduced the additional ending condition that (vi) each object has one of the characteristics in each dimension. In so doing, we accounted for mutual exclusivity and collective exhaustiveness as fundamental ending conditions to satisfy (Nickerson et al., 2013)'s definition of taxonomy. As for subjective ending conditions, we requested that the iterative method converged on a taxonomy that was (i) concise, (ii) robust, (iii) comprehensive, (iv) extendible, and (v) explanatory, as per the lexicon of Nickerson et al. (2013).

Our taxonomy development process involved three iterations. In the first iteration, we used the conceptual-to-empirical approach, where the dimensions and characteristics of the taxonomy are derived (deducted) from existing theories or conceptualisations. We focused on the literature on circular business models and our existing knowledge of this specific domain, motivated by the fact that the logic of transitioning from linear to circular business models is a key antecedent of the sustainable servitization notion and, accordingly, of the offering of sustainable services (Frishammar and Parida, 2019). In other words, when implementing sustainable services, manufacturers leverage circularity practices to align the service business model elements (creation, delivery, and capture of service value) to the principles of sustainability (Rabetino et al., 2024). Our initial taxonomy T1 consisted of dimensions D_1 'CE strategy' with characteristics C_{11} 'close loop', C_{12} 'slow loop' and C_{13} 'narrow loop', and D_2 'business model scheme' with characteristics C_{21} 'ownership' and C_{22} 'non ownership':

$$T_1 = \left\{ \begin{array}{l} \text{CE strategy (Close loop, Slow loop, Narrow loop)}, \\ \text{Business model scheme (Ownership, Non-ownership)} \end{array} \right\}.$$

The characteristics C_{11}, C_{12} and C_{13} were anchored in Bocken et al. (2016). In brief, slowing the loop is about extending a product's life, whereas closing is about closing resource flows after product use. Narrowing focuses on reducing the use of resources use per unit of value, for example through more efficient product use phases. The characteristics C_{21} and C_{22} were derived from, among others, Yang et al. (2018) and Roci and Rashid (2023). Unlike ownership business models, where the provider sells the product to the customer, the provider in non-ownership business models sells the usage or results of the product to the customer and retains ownership (and thus control). Ownership business models are also commonly referred to as product-oriented Product-Service Systems (PSS), while non-ownership business models correspond to use-oriented and result-oriented PSS (cf. Tukker, 2004). We were able to identify multiple instances for each characteristic. For example, by offering after-sales repairs, manufacturers slow resource loops and implement an ownership business model. By offering leasing agreements in which products are remanufactured at the end of each lease cycle, they slow resource loops and implement a non-ownership business model. Since two dimensions were created in this iteration, objective ending condition (iii) was not met, and it was necessary to perform another iteration.

In the second iteration, we applied again the conceptual-to-empirical approach and focused on the general servitization literature as source of additional insights. This led us to revise dimension D_1 as 'service orientation' with characteristics C_{11} 'product' and C_{12} 'customer', and to relabel characteristics C_{21} as 'ownership-based' and C_{22} as 'access-based':

$$T_2 = \left\{ \begin{array}{l} \text{Service orientation (Product, Customer)}, \\ \text{Business model scheme (Ownership-based, Access-based)} \end{array} \right\}.$$

The revised dimension D1 draws from a well-documented categorisation in the servitization literature (cf. Raddats et al., 2019). While product-oriented services (C_{11}) primarily support the proper functioning and management of the product, customer-oriented services (C_{12}) assist customers in improving their processes, activities and actions related to operating the product. In practice, C_{11} merges the 'close loop' and 'slow loop' characteristics of our initial taxonomy and thus makes the taxonomy more concise (subjective condition (i)), whereas C_{12} coincides with the 'narrow loop' strategy. Services in the maintenance field are, however, an exception; while unequivocally corresponding to the 'slow loop' characteristic in T_1, they may correspond to either C_{11} or C_{12} in T_2. In particular, if the service is essentially repair and routine maintenance, then in T_2 it falls under C_{11}; if the service refers to a managed service contract where the supplier provides outsourced product maintenance, it is closer to C_{12}. As for D2, the servitization literature confirms non-ownership as the basis for new mechanisms for value creation (e.g., Vandermerwe and Erixon, 2023). We built upon Fliess and Lexutt (2019: p.61) to rename its characteristics for increased clarity. Since two characteristics were merged in this iteration, objective ending condition (iii) was not met, and it was necessary to perform another iteration.

In the third iteration, we employed the empirical-to-conceptual approach, seeking to ensure that a representative sample of real-life objects (manufacturers' sustainable services) had been examined (objective condition (i)). We conducted a systematic review of the nascent literature on sustainable servitization. We searched in the Elsevier's Scopus database for journal articles in English that contained the keyword combinations ('sustainabl* OR green') AND servit* in the title, abstract, and keyword fields. The search was carried out in March 2025 and included articles published until the end of December 2024, returning 199 unique publications. To ensure a full range of potentially relevant publications, we did not confine the search to journals in specific subject areas and/or filter publications based on journal rankings (e.g., Academic Journal Guide (AJG)'s rating system) (Reim et al., 2015). Titles and abstracts of all articles were screened based on three inclusion criteria. The criteria were: (i) articles had to deal with issues related to servitization and environmental sustainability as main topics; (ii) articles had to address aspects related to the incorporation of sustainability emphasis into manufacturers' servitization strategies; (iii) abstracts had to be sufficiently detailed in terms of article content and key concepts. To ensure consistency when assessing each article, this screening process was conducted by the lead author (Rabetino et al., 2023). 175 papers were dropped at this stage, primarily because they had a marginal (or unclear) focus on sustainability and/or servitization, or they were not concerned with manufacturers integrating sustainability and services into their offerings. For example, Sjödin et al. (2024) was excluded because, in developing a conceptualisation of ecosystem management capabilities for solution providers, it only touches upon sustainability as a macro-trend affecting the automotive and transport industry. Similarly, Schiavone et al. (2022) was excluded because it explores the link between digital servitization and the evolution towards sustainability of manufacturing processes (not of market offerings). Through reading the remaining 24 articles in full, we were able to compile a list (available upon request) of 44 sustainable services that manufacturers may offer. The newly identified objects could be mapped to the dimensions and characteristics in T_2 and did not suggest updates to the taxonomy other than renaming C_{11} as 'lifecycle management' and C_{12} as 'operational efficiency' for better alignment with our meta-characteristic. With:

$$T_3 = \left\{ \begin{array}{l} \text{Service orientation (Lifecycle management, Operational efficiency)}, \\ \text{Business model scheme (Ownership-based, Access-based)} \end{array} \right\},$$

all ending conditions, objective and subjective, were met. No further iterations were necessary.

3.2. Stage 2

In the second stage of our research, we adopted a case approach to provide an empirical assessment of the different categories of sustainable services in our taxonomy and of the associated resource requirements. We investigated how these categories and requirements resonate with practice by analysing the case studies of three manufacturers of industrial equipment that have embarked on a sustainable servitization journey.

The case companies were selected as a result of their participation in a university-industry research hub focusing on servitization and service business innovation. We focused on identifying manufacturers that (i) had been engaged with servitization for several years and had a solid record of providing product-related services, (ii) were actively working on giving emphasis to sustainable services in their offerings, (iii) represented different sub-sectors of the industrial equipment industry. We directed our attention to the industrial equipment industry because business-to-business (B2B) markets have been the context of most servitization studies to date (Naik et al., 2020; Nansubuga and Kowalkowski, 2024), and we sought to provide an opportunity for our case insights to contribute to a cumulative research tradition. Moreover, although the application of our taxonomy is not restricted a priori to the context of B2B, the dominant focus of servitization research on B2B markets inevitably led us to adopt this perspective when developing our taxonomy. Our sample of three companies is large enough to allow to assess the consistency between service characteristics noted empirically and those designated within our taxonomy.

Our pre-understanding of the companies through prior collaboration in several research projects made it possible to focus our data collection on the core issues, relying on a single informant per company. We tackled the potential bias introduced by the use of a single respondent through carefully selecting the informants (Vachon, 2005) and ensuring that they had the necessary knowledge to provide rich and detailed information about their company's service activities. The informants were selected because they were actively involved in implementing sustainable servitization initiatives. All of them were executives who had multi-year experience in leading the delivery and management of servitized offerings. The data collection was conducted in form of semi-structured interviews supported by an interview guide. The guide was developed based on themes relating to the description of sustainable service offerings, needs and practices related to their implementation, and critical knowledge/resource challenges. Follow-up questions were extensively used to clarify points and obtain further details, allowing to capture interesting insights as they emerged (Eisenhardt, 1989). The interviews lasted from 60 to 90 min and took place via online conferencing. Interviews were recorded and transcribed, and the transcripts provided the basis for the data analysis. Interview data was supplemented with secondary information from company websites, online news articles and press releases – which was collected with the aim of getting a more complete view of the sustainable services offered, as well as to triangulate and validate the interview data (Yin, 2014).

The analysis of interview transcripts was based on thematic analysis (Braun and Clarke, 2006). We followed a process of deductive coding, whereby the four service categories in our taxonomy were used as predetermined codes to classify text. Where passages or sentences referred to multiple service categories, they were classified within multiple parallel codes. Because our focus included resource aspects, we highlighted the text that appeared to describe resources needed to offer the service categories, and sub-coded it thematically into five resource codes (knowledge, physical, relational, organisational, information – illustration will be provided in the next section), while also remaining open to unexpected codes emerging from the data. The process was performed as a within-case analysis for each case company, followed by a cross-case analysis where the companies were compared.

4. Study findings

4.1. Taxonomy of manufacturers' sustainable servitization approaches

As previously stated, throughout this study we use the concepts of sustainable servitization approaches and categories of sustainable services interchangeably. Moreover, we focus exclusively on product-related services (i.e., services that deliver value that is tied to a manufacturer's product offerings) (Mathieu, 2001b; Baines and Lightfoot, 2014) and exclude other 'non-good services' (cf. Judd, 1964: p.59) that manufacturers may offer such as banking, staffing, general consulting, waste management and remediation. As illustrated in section 3.1, our taxonomy of manufacturers' sustainable services has two dimensions. The first dimension is the 'lifecycle management' or 'operational efficiency' orientation of the services; i.e., services are aimed at either extending the product lifespan and closing resource loops or at enabling more efficient operating patterns. The second dimension is the business model; i.e., sustainable services may be provided according to an 'ownership-based' or an 'access-based' scheme. The latter would be the case of sharing, renting, leasing, product-as-a-service, and similar access-based business models, where manufacturers sell product access and results instead of ownership (Benedettini, 2025). The taxonomy therefore covers also those manufacturers that are adopting CE practices relevant to slowing and closing resource loops (e.g., repair, technical upgrade) to reduce the product-related costs that they internalise as a consequence of retaining product ownership (service category 2). It similarly classifies the offers of those manufacturers that are adding operational efficiency services as a way to support customers in increasing the operational efficiency of running the 'accessed products' (cf. Ackermann and Tunn, 2024), ranging from technical advice and training to full management of product-related activities (service category 4). Although much of the discourse around sustainable servitization in the CE literature concentrates on business models where manufacturers retain ownership during the product lifecycle (Menon et al., 2024; Rabetino et al., 2024), the broader servitization research (e.g., Korkeamäki et al., 2022; Raddats et al., 2022) has shown it to be quite common with manufacturers providing not only lifecycle management services, but also operational efficiency services, as add-ons to products sold to customers (categories 1 and 3, respectively). Hence, four categories of manufacturers' sustainable services are proposed, which are based on the four quadrants in Fig. 2. These are not mutually exclusive; that is, a manufacturer's sustainable servitization strategy may include multiple categories of sustainable services (Burton et al., 2017; Heirati et al., 2023). For example, a manufacturer of industrial equipment may offer service packages that include refurbishing and reconditioning of products sold to the customers (quadrant 1), while also implementing leasing programmes where products are remanufactured at the end of each lease cycle before re-leasing (Mutha et al., 2022; Gao et al., 2023) (quadrant 2). Other examples of services that fall within each category are provided in Fig. 2. These categories or approaches are analysed individually hereafter; an analysis is also provided of the key resources that enable their implementation by product manufacturers (Fig. 3).

4.1.1. Lifecycle management services with ownership-based business model

Traditionally, service offers that transfer product ownership to the customers have been assumed to inherently prioritize the generation of revenues from continued sales of new product units over providing lifecycle services to prolong product life and avoid resource depletion (Huikkola and Kohtamäki, 2017; Stabler et al., 2024). An often-heard claim is that these approaches encourage a focus on selling as many products as possible (Tukker, 2004), leading manufacturers to offer only those lifecycle services that customers normally expect (e.g., repair, routine maintenance) or that are legally demanded (e.g., take-back) (Kumar et al., 2024; Stabler et al., 2024), so as to avoid

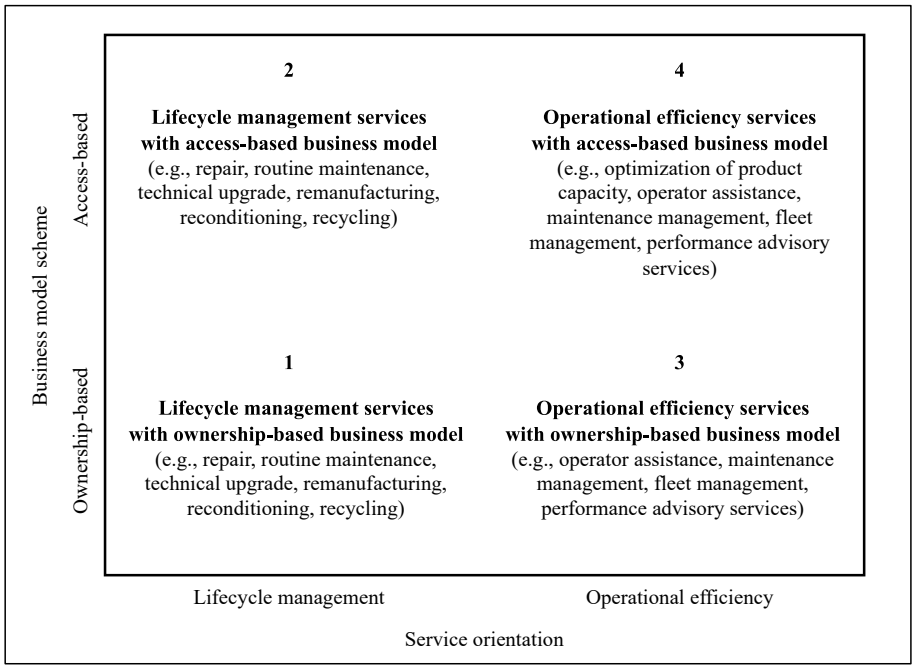

Fig. 2. – Taxonomy of manufacturers' sustainable services.

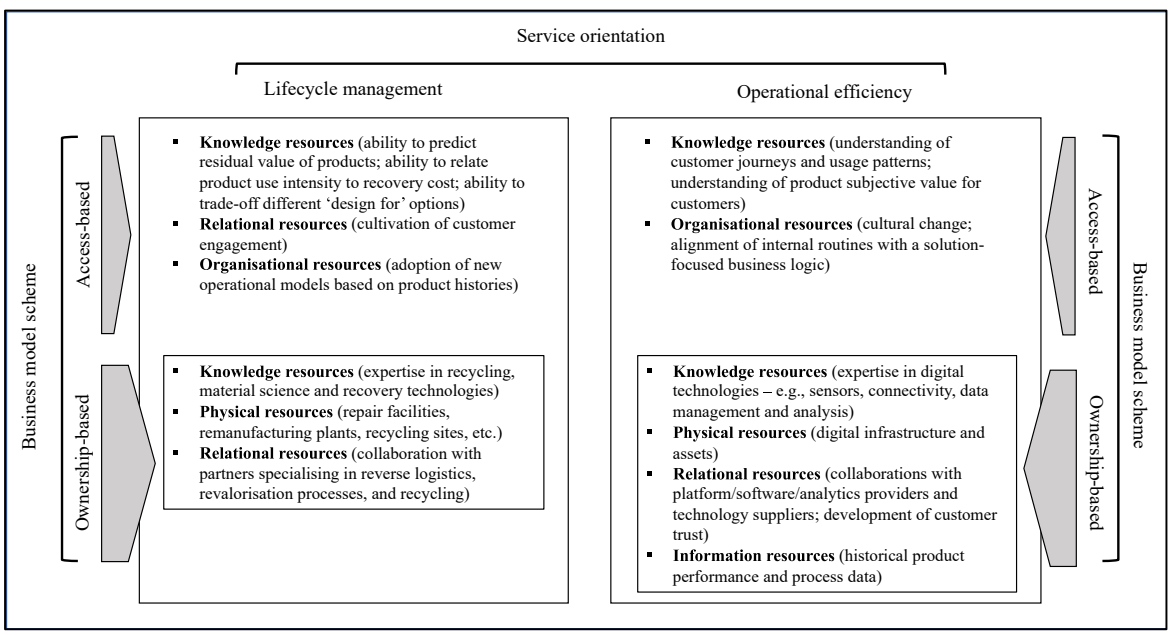

Fig. 3. – Key resources for the implementation of each service category.

cannibalisation of core revenue streams from product sales (Atasu et al., 2010). However, a detailed examination of the literature suggests multiple reasons why manufacturers may decide to provide extended lifecycle management services for products that are owned by customers. First, if they do not provide such services, chances are independent third-party competitors will, cannibalising market shares from new products while not opening new revenue streams from services in return (Atasu et al., 2010; Mutha et al., 2022). Second, by adding lifecycle management services to conventional product offerings, manufacturers may target new sales to environmentally conscious customers (Kumar et al., 2024; Stabler et al., 2024) or navigate pressures from low-priced or new competitors' products (Pearce, 2009; Atasu et al., 2010). Third, supply chain resiliency can be bolstered in dynamic environments via increased material self-sufficiency (Chowdhury et al., 2025). Finally, lifecycle management services can provide profit margins even two or three times higher than product sales (Pearce, 2009; Kowalkowski et al., 2017).

In terms of critical resource enablers (Fig. 3), planning and organising the offering of these services is likely to require a combination of ***knowledge resources***, ***physical resources***, and ***relational resources***. Since recovery operations necessitate high levels of product knowledge and expertise (Mutha et al., 2022), original product manufacturers have an obvious advantage over third-party providers in implementing services such as refurbishing, remanufacturing, reconditioning, and upgrading (Pearce, 2009). However, they still need to gain *knowledge* of relevant engineering technologies (e.g., recycling, material science) and

to develop specialised skills for better utilising technological advances to restore and recover end-of-use products (Zhou et al., 2024). In addition to committing to investing in technological competences (knowledge resources), manufacturers need to establish repair facilities, remanufacturing plants, recycling sites, etc., with significant investment in *physical assets* (Kohtamäki et al., 2024). Nevertheless, *relational resources* are likely to be important. Handling the reverse flows underpinning service activities that result in slower and more closed resource loops often requires the involvement of multiple network actors to bring about the necessary qualifications, expertise and capabilities (Spring and Araujo, 2017; Frishammar and Parida, 2019). Relational resources allow manufacturers to collaborate with selected partners that specialise in reverse logistics, revalorisation processes and recycling (Rabetino et al., 2024).

4.1.2. Lifecycle management services with access-based business model

As previously mentioned, sustainable services where manufacturers retain product ownership refer to offerings such as sharing, renting, leasing, and product-as-a-service. Ownership retention allows for greater control over the product and motivates manufacturers to recover value at the end-of-life through reuse, remanufacturing, reconditioning, recycling, and other closing loop strategies (Frishammar and Parida, 2019; Kühl et al., 2020; Sun et al., 2025). It also sets incentives for prolonging the life of the product through better (e.g., more frequent) maintenance and repairs because manufacturers may look to maximise their profits based on enlarged product longevity and reduced failure rates (Kühl et al., 2020; Kanatlı and Karaer, 2022; Rabetino et al., 2024). While maintenance and advanced end-of-life recovery are not marketed as optional 'for fee' services (as with conventional ownership-based business models), they are usually assumed to be an inherent part of the non-transfer of product ownership to the customer (Sundin and Bras, 2005; Stabler et al., 2024). This is because resource efficiency and waste reduction align with the key aspect of profitability for the manufacturer, and therefore it is in the latter's interest to ensure the optimal lifecycle management of the product. Moreover, customers are more willing to accept used and second life products because they do not have to buy them (Yang and Evans, 2019). Finally, the manufacturer gets the opportunity to leverage the environmental benefits of extensive application of circular practices to achieve a distinctive (sustainability-oriented) corporate image (Johl et al., 2024; Kumar et al., 2024). Such incentives related to product recovery are commonly believed to have cascade effects on design practices. It is suggested that manufacturers will design products for durability, repairability, remanufacturing, upgrading and recycling – that is, to facilitate life extension and value retention processes (Spring and Araujo, 2017; Kanatlı and Karaer, 2022; Johl et al., 2024; Chowdhury et al., 2025; Stabler et al., 2024). Hence, it can be expected that higher environmental value is created than if product ownership was transferred to the customer.

In terms of critical resource enablers (Fig. 3), the configuring and management of this category of sustainable services requires further ***knowledge resources***, ***relational resources*** and ***organisational resources*** in addition to the knowledge, physical and relational resources previously discussed in relation to providing lifecycle management services with an ownership-based business model. Indeed, necessary *knowledge* is likely to include the ability to predict the residual value of products after multiple use cycles, so as to assure adequate pricing and terms of rental/access contracts (Fallahi et al., 2023). Similarly, suppliers need to be able to identify threshold values of use intensity that make it environmentally friendly and financially profitable to restore and lease again the product at the end of a use cycle (Gao et al., 2023), as well as to assess product and market characteristics that may cause a misalignment between economic and environmental goals (Örsdemir et al., 2019; Sun et al., 2025). Nevertheless, manufacturers need to understand the trade-offs between the different relevant 'design for' options (Rabetino et al., 2024). Additional *relational resources* concern the cultivation of customer engagement, which is necessary to lead customers to treat products responsibly and prevent accelerated wear and shortened lifespans (Stabler et al., 2024), besides increasing normative acceptance of recovered products (Frishammar and Parida, 2019). While not affecting the physical resource requirements for recovery activities, retaining ownership requires manufacturers to anchor their product management routines on product histories, and thus to possess adequate *organisational resources* to adopt a new operational model (Kohtamäki et al., 2024).

4.1.3. Operational efficiency services with ownership-based business model

With servitization approaches focusing on operational efficiency services, the goal of creating environmental value concerns the product use phase, and hence the customer's domain (Rabetino et al., 2024). Even though product ownership is transferred to the customer, the manufacturer becomes part of customer processes related to operating and deploying the product (Sjödin et al., 2023). This generally implies that provider and customer co-create value in the customer's use situation through mutual resource integration, combining their activities and interacting in a joint sphere to ensure customer success (Kowalkowski and Ulaga, 2024). Environmental value stems from enabling more eco-friendly operating patterns and typically encompasses enhanced energy/resource efficiency and reduced emissions. For example, manufacturers may offer customised training on how to operate products efficiently (e.g., driver training regarding use of gears and brakes to reduce fuel consumption and extend the life of trucks) (Frishammar and Parida, 2019; Kühl et al., 2020, 2022) or over-the-air software updates to secure the optimal performance and longevity of the product (Stabler et al., 2024). They may also offer maintenance contracts where product tracking, predictive analytics and data-driven insights allow strategies focused on optimising maintenance schedules, avoiding unnecessary maintenance actions and preventing needless replacements of components (Sjödin et al., 2023; Opazo-Basáez et al., 2024; Stabler et al., 2024). Other examples of operations optimisation services include fleet management or site management services, which provide industrial customers with analytical support to optimise the use of vehicles in their production sites (e.g., route planning, loading, operating speeds), reducing fuel/energy consumption, product wear, and emissions (Frishammar and Parida, 2019; Sjödin et al., 2023). Similarly, performance advisory services broadly focus on helping customers uncover inefficiencies and opportunities for more sustainable product usage (Sjödin et al., 2023). As these examples illustrate, operational efficiency services normally entail a higher degree of customisation and relational interaction than lifecycle management services.

Accordingly, in terms of critical resource enablers (Fig. 3), implementing this approach to sustainable servitization demands a different set of ***knowledge resources***, ***physical resources*** and ***relational resources***, as well as specific ***information resources***. Operational efficiency services are typically based on continuous monitoring and analysis of product activity, so as to leverage data-driven insights for enhanced decision-making (Sjödin et al., 2023; Opazo-Basáez et al., 2024) and evolve service offerings based on customer feedback. Competences in digital technologies, such as sensors, connectivity, data management and analysis, are therefore fundamental *knowledge resources* for providers (Davies et al., 2023; Sjödin et al., 2023). Leveraging data about product condition and use to support customer processes and internal service operations also requires implementing a strong digital infrastructure (Sjödin et al., 2022; Opazo-Basáez et al., 2024). Installing, maintaining and continuously upgrading this infrastructure, which will need to manage complex networks of connected products, is not feasible without investments in physical resources (Tronvoll et al., 2020; Benedettini, 2024). Relatedly, both in academic literature (e.g., Sklyar et al., 2019; Tronvoll et al., 2020; Momeni et al., 2023) and industry reports (e.g., Systemiq, 2023; Stena Circular Consulting and Cradlenet, 2023), it is frequently emphasised that manufacturers rely significantly on dedicated partnerships and collaborations with external actors (such as platform providers, software providers, data analytics providers, and

technology suppliers) to develop and deliver their operational offerings. This gives rise to the need for *relational resources* to build trustful relationships with such actors (Kolagar et al., 2022), align incentives, and encourage them to invest in relation-specific assets (Kamalaldin et al., 2020). Nevertheless, relational resources are important to gain customer trust and increase customers' willingness to share product and usage data for deriving operational decisions that maximise efficiency in their unique situations (Kolagar et al., 2024; Madanaguli et al., 2024). Finally, to enable proactive and well-informed decisions regarding how to optimise material and energy use, manufacturers need *information resources* in the form of product performance and process data collected from an existing installed base (Ulaga and Reinartz, 2011; Sjödin et al., 2023).

4.1.4. Operational efficiency services with access-based business model

In the field of operational efficiency services, the retention of product ownership by the manufacturer enables solutions that facilitate sharing and sufficiency (Stabler et al., 2024). The goal of reducing the environmental impact associated to the product use phase is pursued by allowing multiple customers to access the same product sequentially, thereby increasing product utilisation and reducing the overall number of product units that need to be in use to meet the market demand (Frishammar and Parida, 2019; Reim et al., 2015; Bond et al., 2020; Abdelkafi et al., 2022; Kanatlı and Karaer, 2022). Providing access and usage to products instead of ownership also allows manufacturers to offer product configurations that match changes in the needs and use situations of individual customers (Kowalkowski and Ulaga, 2024), sizing the products appropriately for the lowest resource and energy consumption (Benedettini, 2025). Moreover, since the higher product cost is borne by the manufacturer, it becomes easier to introduce digital technologies into customers' operations, and thus extensively apply analytics to in-use and historical data for enhancing resource utilisation and efficiency (Menon et al., 2024). At the same time, manufacturers are incentivised to 'design in' connectivity and smart functionalities to their products, while also adopting product designs that incorporate the ability to accommodate developments in sensor and software technologies (Porter and Heppelmann, 2014). The internalisation of maintenance, underutilization, and other product operating costs incentivises manufacturers to aim for the highest possible use phase efficiency (Kanatlı and Karaer, 2022) – especially if they also offer to operate the product on behalf of the customer and thus take on the cost of fuel or energy to run the product (Mutha et al., 2022). Clearly, as with lifecycle management services, the shift from customer ownership to provider ownership drives improvements to the environmental value creation potential of the services.

In terms of critical resource enablers (Fig. 3), the pool of knowledge, physical, relational and information resources that provide the foundation to implement operational efficiency services on customer-owned products are a necessary requirement even when product ownership stays with the provider. However, it can be expected that retaining product ownership, along with the abovementioned incentives to be as resource efficient as possible in providing the product utility, introduces the need for manufacturers to possess additional ***knowledge resources*** and ***organisational resources***. Indeed, necessary *knowledge* likely includes also a deep understanding of customer journeys and usage patterns, in combination with an appreciation of the subjective value that individual customers derive from product adoption and use (Sjödin et al., 2023). Without such understanding, manufacturers may end up jeopardising their efforts to maximise product utilisation and size products appropriately to customers' actual needs. Nevertheless, this approach to sustainable servitization involves far-reaching changes in multiple organisational dimensions, as manufacturers need to align their operational model and internal routines with a solution-focused business logic. This alignment process entails cultural change (Frishammar and Parida, 2019; Huikkola et al., 2022) and requires adequate *organisational resources* to orchestrate the necessary reconfiguring of value-chain activities for improved product use, greater service orientation, and customer centricity.

4.2. Empirical assessment

An overview of the three (anonymised) cases that were used to empirically assess our taxonomy is provided hereafter. **MachineTool** is a supplier of precision manufacturing technologies, including milling, EDM (Electrical Discharge Machining), laser texturing, laser micromachining, and additive manufacturing. It provides machines, automation solutions and services to manufacturers of high-precision parts, tools, and mold-making in key industries like aerospace, automotive, and medical. It has a presence in over 50 countries worldwide, with sales companies in 25 countries and production sites in eight locations across Europe, the Americas and Asia. **Power** is a provider of electrical equipment intended to offer power availability, control, and safety for low-voltage electrical networks. Its portfolio includes UPS (uninterruptible power supply), switching devices, energy metering systems, and energy storage solutions, as well as software and services. With around 30 subsidiaries located on all five continents, it serves customers in critical sectors like healthcare, industry, and data centres ensuring service continuity and boosting energy efficiency. **Cooling** specialises in precision air-conditioning and humidity control solutions for mission-critical applications, particularly data centres and telecommunications. It offers air handling units, micro data centre, chiller units, monitoring systems, and other related equipment and services, allowing customers to choose between standardised solutions or fully customised designs. It maintains a global network of subsidiaries, production sites, and sales and service partners, through which it ensures local support and expertise in over 140 countries. All key informants were from the Italian subsidiaries of the companies.

For each case company, a particular approach to sustainable servitization is dominant. Lifecycle management services with ownership-based business model (quadrant 1 in Fig. 2) represent the main category of sustainable services in the MachineTool case. In the Power case, the main offering also refers to an ownership-based business model but concerns operational efficiency services (quadrant 3). Although all three companies in some cases keep the ownership of the products themselves, only in the Cooling case does the sustainable servitization strategy embrace product-as-a-service schemes and incorporate the shift to an access-based business model as a main theme. However, such transformation initiatives were recently launched, and Cooling focuses its attention on enhancing the product's operational efficiency and intensity of use (quadrant 4) rather than on pursuing more sustainable lifecycle management (quadrant 2).

4.2.1. Lifecycle management services with ownership-based business model

MachineTools performs revamp and refurbishment work to extend the lifespan of machines and parts, take-back of end-of-life machines for reconditioning of components and recycling of steel and cast iron sections, and collection and recycling of a particular consumable wire used by its EDM machines. Power performs retrofit of spare parts and provides an end-of-life disposal service for the batteries of its UPS systems. It also offers a lifecycle extension service based on its UPS having a modular architecture that allows the operational life to be extended by hot-swap replacement of obsolete electronic modules. Moreover, all companies investigated offer repairs and routine maintenance. The interviewees noted that the provision of these services by their companies builds on a major shift that has occurred in product designs, with digital control systems and technical modularity becoming established as common design features of business-to-business equipment and products. For example, the information stored in the CNC (computerised numerical control) unit is vital for MachineTool to evaluate the technical and economic feasibility of revamping a machine because it allows to estimate the likely degradation of components by tracking down what has happened during the operating life. Similarly, the evolution from

monolithic to modular UPS has been a prerequisite for Power to develop its lifecycle extension concept; i.e., a product architecture with electronic-free cabinet and plug-in modules.

For the companies under study, providing services in this category requires primarily knowledge and physical resources. Despite holding deep product knowledge, they lacked experience in second-life technologies and technical equipment for other than repair activities. To obtain these resources faster and speed up the implementation of the services, MachineTool acquired a company that specialised in machine refurbishments and rebuilding of components and devices, whereas Power opted for internal development. Relational resources appear less important, as the companies' external collaboration is limited to specific logistics and recycling activities (e.g., transportation of machines from/to refurbishing and revamping sites, recovery of raw materials). Indirectly, these insights provide an indication of which resources it is important to develop internally or purchase (and, thus, own) and which can be simply accessed through outsourcing and business partnering.

4.2.2. Lifecycle management services with access-based business model

Power offers a leasing formula that includes both rental and maintenance of UPS systems. MachineTool provides full-service rental and maintenance options for EDM machines. The leasing/rental contracts of both companies normally cover a period of several years. In contrast, Cooling offers short-term rental (mostly for emergency or temporary use) and is currently developing a cooling-as-a-service scheme. For all three case companies, these offers currently account for a very minor portion of the business. Power and MachineTool have realised that very few customers are interested in leasing/rental arrangements and hence the non-transfer of product ownership does not incentivise a stronger pursuit of product longevity and recirculation on their part. The cooling-as-a-service initiative has required Cooling to make some changes to the design of its equipment so that it is easier and quicker to transfer from customer to customer. While these changes create environmental value by preserving the equipment durability, Cooling does not plan to make other significant investments into extending the longevity of equipment of which it retains ownership or into recovering value at the end-of-life. The main reason for this is that the majority of the equipment total cost of ownership consists of energy-related expenses; hence, pursuing enlarged equipment longevity and end-of-life recovery presents a less attractive avenue than energy optimisation for delivering cost savings to customers and/or profiting from the internalisation of equipment-related costs.

Despite limited, the experience of the investigated companies confirms that ownership retention makes it more resource-demanding to provide lifecycle management services, especially in terms of knowledge and relational resources. Lacking appropriate knowledge, Cooling needed to hire new personnel to be able align the equipment design with the new transportability and durability requirements. Similarly, strong ties with customers emerge as an important relational resource since Cooling is leveraging supplier-customer integration to take charge of a range of activities in the operational environment that affect maintenance requirements and, therewith, the durability of its equipment. The case companies did not report the need for organisational resources; however, since they only provide services in this category to a very limited extent, they have not attempted so far to adapt their product management routines to the new business model.

4.2.3. Operational efficiency services with ownership-based business model

Power offers condition-based maintenance, natively included in the service contracts of its new generation UPS systems. This maintenance strategy makes it possible to limit replacements of consumable components and on-site maintenance visits to only essential actions, reducing component waste and carbon footprint from technicians' travels. Power also increasingly executes maintenance inspections remotely, removing the need for physical travel. All case companies offer advice and consultancy services in order to support customers in improving the operational efficiency of their equipment. Along these lines, they have also developed web-based platforms that make real-time monitoring, control, and optimisation of equipment status and performance available to customers, helping them boost uptime, energy efficiency and reliability. The informants emphasised that their companies' customers place utmost value on the cost savings and trouble-free operations delivered by services in this category, but (with the exception of some large customers that are committed to sustainable procurement) they are not particularly attracted by sustainability-related benefits. Informants also discussed how the evolution of the equipment industry towards smart and connected products, which enables access to product usage data, is the backbone for providing lifecycle management services.

In this respect, the inclusion of built-in digital features in many current equipment designs allows the companies investigated to easily manage the sensor and connectivity challenges of offering such services (in some cases, it was not even necessary to add sensors to the equipment). Skills in data analytics and artificial intelligence emerged as a key enabler of the services, despite not a traditional area of strength for the companies. Moreover, intrinsic to the shift to condition-based maintenance is that Power takes responsibility for monitoring whether the equipment is working or not; its ability to do so depends on a reliable and high-performing cloud infrastructure, and thus on physical resources. To get access to data analytics competences and cloud infrastructure, Power exploits dedicated partnerships with digital consulting companies, universities and platform providers. While allowing Power to keep up with rapid technology changes and scale its services across broader markets, these partnerships require relational resources. In contrast, MachineTool and Cooling have so far been able to incorporate the (more limited) software and analytics capabilities they needed through the acquisition of specialised providers or new personnel, which allows them to be less dependent on relational resources. However, relational resources are a fundamental precondition for all three companies to engender customer trust and obtain access to product usage data despite potential privacy, security, and AI ethics issues. In this respect, the respondents emphasised that, unlike product usage data (which belong to the customer), installed base data about product failures, consumption patterns, performance problems etc. are abundantly available to most manufacturers because new information is constantly acquired during maintenance activities. Therefore, despite its relevance to operational efficiency services, this particular information resource does not require specific efforts – it simply grows richer as maintenance records are accumulated.

4.2.4. Operational efficiency services with access-based business model

With its rental and cooling-as-a-service schemes, Cooling focuses on enabling customers to align equipment capacity to changes in their operational needs, avoiding that cooling systems are left unused because, for example, over time they become undersized or oversized. These offers have maximum economic and ecological impact for customers, since particular emphasis is placed on improved energy efficiency and the utilisation of power capacity. Similarly, leasing contracts incentivise Power to adopt condition-based maintenance strategies because they are provided in collaboration with a financial partner, with a fixed fee paid to Power for guaranteeing equipment efficiency and performance through servicing. Accordingly, similar to when an ownership-based business model is adopted, offering services in this category resorts to product connectivity, data analytics, digital infrastructure, access to usage information, and availability of installed base data.

Moreover, Cooling confirmed the need to deploy additional knowledge as well as organisational resources. Specific knowledge resources have been necessary to capture the individual needs of customers. Avoiding that the customers keep a large number of out-of-service units allows to easily capture the 'low hanging fruits'; yet, in order to maximise the energy efficiency and ecological footprint of climate solutions, it is fundamental to have the competence to view the cooling system

holistically and to understand how multiple units work together in the customer environment. Organisational resources so far have been especially important to establish a separate sales unit for rental and cooling-as-a-service offerings, so as to create a necessary focus on these offerings within the sales department and ensure that the development of the new business model was not hampered by the dominant perspective of selling equipment and services.

5. Discussion

As rising environmental and climate concerns create the need to embed sustainability more centrally into business value propositions, the strategic logic of servitization and the ecology-oriented CE share increasingly converging goals and practices (Spring and Araujo, 2017; Kohtamäki et al., 2024). We address the lack of clarity regarding the umbrella concept for this emergent phenomenon (Rabetino et al., 2024), that is, sustainable servitization and provide a more nuanced characterisation of what it entails than extant literature.

Uncovering the heterogeneity of possible approaches to sustainable servitization, we propose a taxonomy that classifies them into four distinct categories based on service orientation (lifecycle management or operational efficiency) and business model scheme (ownership-based or access-based). The literature shows a variety of service categorisations concerning the general field of servitization (for a review, see Benedettini and Neely, 2018; Raddats et al., 2019). In most cases, these consist of two dimensions, one of which is frequently the product versus customer orientation of the service (e.g., Oliva and Kallenberg, 2003; Raddats and Easingwood, 2010; Ulaga and Reinartz, 2011; Coreynen et al., 2017). Our taxonomy suggests that, in the specific field of sustainable servitization, this dimension can be effectively reinterpreted to classify product-related services as supporting lifecycle management or supporting operational efficiency. Moreover, by also drawing upon the literature on circular business models, our taxonomy development captures that categorising sustainable services based on orientation towards lifecycle management or operational efficiency largely reflects the notions of closing, slowing and narrowing resource loops (Bocken et al., 2016; Geissdoerfer et al., 2018). In line with Bocken et al. (2016), we identify common ground in the approaches of closing and slowing resource loops, with lifecycle management services using either one or the other to generate environmental value. Yet, while Bocken et al. (2016: p. 310) argue that narrowing 'does not involve any service loops', we show that industrial manufacturers today offer several (operational efficiency) services in this area. In addition, our taxonomy links to other categorisations in the servitization field which, building on the business model framework, focus on the nature of the value proposition/revenue model. However, we propose that, despite most of these categorisations consider whether the service value proposition is input-based or output-based (e.g., Ulaga and Reinartz, 2011; Kindström and Kowalkowski, 2014; Baines and Lightfoot, 2014; Coreynen et al., 2017), the key business model differences between sustainable services depend on whether product ownership is transferred to the customer or not. In so doing, our taxonomy aligns with Matthyssens and Vandenbempt (2010)'s framework, which categorises service offerings based on whether they are provided as an add-on to products or as an integrated service package that also includes products.

Moreover, in addition to illustrating the different ways in which manufacturers can generate environmental value for their customers and society via the service dimension, we highlight the unique resource requirements they must be ready to fulfil. We show that the critical resources vary across different categories of services, in line with previous studies such as Ulaga and Reinartz (2011). While it seems evident that realising and implementing sustainable services (regardless of the category) requires the combination of multiple resources, our research shows that the use of digital technologies is a major precondition for many operational efficiency services and, indeed, we identify several types of digitalisation-related resources as critical to the provision of such services. Thus, we corroborate the general claim in previous studies, such as Sjödin et al. (2023), Kolagar (2024), Kolagar et al. (2024) and Gebauer et al. (2025), that sustainability-oriented servitization and digital servitization are deeply intertwined, but at the same time we point out that this view predominantly reflects the context of operational efficiency services.

5.1. Theoretical contributions

This study provides three main theoretical contributions. First, we point out a definitional problem in the extant literature concerning the notion of sustainable servitization, with different or ambiguous interpretations given to the underlying concept of combining environmental sustainability with service-based value creation in the business strategies of product manufactures. This means that existing studies offer scattered and hardly comparable results, and the risk exists that theoretical advances become lost in different interpretations that hamper knowledge accumulation. By exposing the problem, we underscore that sustainable servitization remains a theoretically undeveloped concept compared to other terms such as (traditional) servitization (e.g., Momeni et al., 2023; Chirumalla et al., 2023) or digital servitization (Kohtamäki et al., 2019; Sklyar et al., 2019; Tronvoll et al., 2020; Benedettini, 2024). Until conceptual consolidation is achieved, it will be difficult to integrate insights from different studies or research areas beyond the surface level, and cumulative research progress is likely to proceed slowly.

Second, in contrast to a common assumption in the literature suggesting that sustainable servitization is a generic strategy, our taxonomy unpacks the heterogeneity of the sustainable services that product manufacturers provide to their customers; these services differ in terms of their value creation mechanisms, operational processes, and critical resource enablers. In the service domain, taxonomies help address the complexities of services by identifying criteria that reflect core service attributes (Ng et al., 2007), assist managers in developing meaningful strategies (Sampson and Froehle, 2006), and provide a foundation on which researchers can build theory (Cook et al., 1999). Taxonomies and other classification schemes have been of great utility for servitization practice and theory (e.g., Mathieu, 2001b; Oliva and Kallenberg, 2003; Ulaga and Reinartz, 2011; Baines and Lightfoot, 2014). However, taxonomies concerning the sustainable services with which manufacturers complement their product offerings are virtually non-existent, and the few general classifications that are available for sustainable services (e.g., Cocca and Ganz, 2015) are only partially applicable to servitization contexts. By proposing our taxonomy and unpacking each of the four service categories identified, we provide granular insights into specific sustainability tactics and associated resource requirements. Without this granular view, researching sustainable servitization can only lead to simplistic and inaccurate results.

Third, our findings challenge conventional assumptions about access-based business models. While the adoption of such models is typically considered to be foundational and implicit to sustainable servitization (e.g., Agrawal and Bellos, 2017; Örsdemir et al., 2019; Kühl et al., 2022; Rabetino et al., 2024), our findings suggest that it is not an essential condition for generating environmental value through services. Specifically, we find that, for both lifecycle management and operational efficiency services, conventional ownership-based models play a more important role for our case companies. However, this does not imply that product ownership by the manufacturer is becoming less necessary to unleash the full environmental potential of servitization. Prior research highlights that ownership retention provides a business case for investments that improve product endurance, recirculation and utilisation, since the revenue model does not primarily reward the number of products sold (Frishammar and Parida, 2019; Kühl et al., 2020; Kanatlı and Karaer, 2022). However, the executives interviewed acknowledged that the business case is typically not as strong as conceptual literature would suggest. Environmental sustainability alone is

not enough for customers; manufacturers need to ensure also other benefits (e.g., cost savings, flexibility, innovation) when introducing environmental orientation in their service value propositions, which impairs the ability to capture the profits expects from product ownership. Thus, manufacturers tend to be more receptive to the opportunities offered by traditional ownership-based models, just as they have been found to focus their general and digital servitization strategies on discrete offerings (product sales and base/intermediate services; cf. Baines and Lightfoot, 2014) rather than advanced offerings with capabilities delivered through product use (Kowalkowski et al., 2015). In this context, our study also points to the issue that, despite its significant environmental appeal, access-based business models are 'not for everyone' (Pearce, 2009): manufacturers need to have the appropriate resources and expertise. The greater resource requirements associated with more complex sustainable services introduce core rigidities and act as barriers that mitigate the practical relevance of such services.

5.2. Managerial contributions

This study makes two main managerial contributions. First, many manufacturing companies are still in early stages of applying a sustainability perspective to their servitization strategies and integrating circularity principles into their service offers. Against this backdrop, the taxonomy developed in this study can help them think through how to extend and enhance extant service portfolios in ways that contribute to environmental sustainability. By considering the four characteristics that define our taxonomy, managers of manufacturing companies can get a clear picture of the intrinsic differences among different types of sustainable services and make a reasoned judgement about how to structure sustainable servitization initiatives. Our taxonomy provides support to making decisions such as what service options are most appropriate to extend the existing offering portfolio in line with the company's sustainability goals, what settings are necessary to ensure the adequate provision of specific sustainable services, or what the opportunities are to evolve sustainable service offerings over time.

Second, our findings communicate the importance for managers to be cognisant of the need to activate specialised competences and resources when adopting and implementing a sustainable services orientation. Insights gained from our study provide guidance to examine which resources (e.g., physical, knowledge, organisational) are already in place, and which need to be built. Next, managers should decide how to best obtain the newly required competences and resources. The interviewed companies looked outside their boundaries. However, they generally favoured acquiring smaller companies that already possessed specialised expertise or hiring employees/consultants with key necessary skills, rather than leveraging external resources through partnerships, strategic alliances and other collaborative agreements involving co-provision of sustainable services with technology or third-party providers. Thus, in a way, our results point to vertical integration as a generally more appropriate organisational form than externalisation for embracing sustainable servitization.

5.3. Limitations

This study has its limitations. First, the proposed taxonomy and service categories were inferred from the literature, restricted to service attributes represented in existing articles and to service examples cited therein. Moreover, given the rapid pace of digital technology development and the digital nature of many lifecycle management services, it is likely that new opportunities for services aimed at delivering environmental value will emerge in the future. Hence, while our taxonomy provides an initial framework for analysing manufacturers' sustainable services, it should be reviewed from time to time to account for new or currently underrepresented service types. Second, we do not contend that our suggested resource requirements represent an exhaustive list, but rather a contributory collection that we provide with the purpose of encouraging future research to delve deeper into the preconditions that manufacturing companies face when implementing different categories of sustainable services. Despite our effort to cross-check our findings with three case companies, the identified taxonomy of enabling resources needs further empirical validation with a larger sample. Third, the study did not measure the performance or success of companies providing the four categories of service offerings, nor we do provide an indication of the relative importance of different relevant resources in terms of delivering environmental and business value. While our analysis of resource enablers provides an indication of the general viability of the service offerings, it would be useful to offer a quantification of the effectiveness of those offerings and the fit with (firm-/industry-level) context conditions. Despite these limitations, this study contributes by providing a more fine-grained understanding of the notion of sustainable servitization and theorising about its heterogeneous nature. We hope that our work will stimulate further research that focuses on sustainable servitization and continues developing theory in this increasingly important research domain.

CRediT authorship contribution statement

Ornella Benedettini: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Federico Adrodegari:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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